

ME 2911 – MAE Practice II

Temperature Sensor Test Rig

Standard Operating Procedures

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1 Overview

The test rig shown in Figure 1 is used to test the dynamic response of different temperature sensors, including four different thermocouples and one RTD. The hair dryer is attached to the right-hand side of the test rig with the purpose of generating hot air flow. Two gate valves are located to the left of the hair dryer with the purpose of directing air flow. Adjusting the gate valves can direct air down the tube towards the temperature sensors or through a bypass that exhausts to the room. The DAQ, located on the left-hand side, measures and collects data from the temperature sensor that is then written into a data file using the LabView interface.

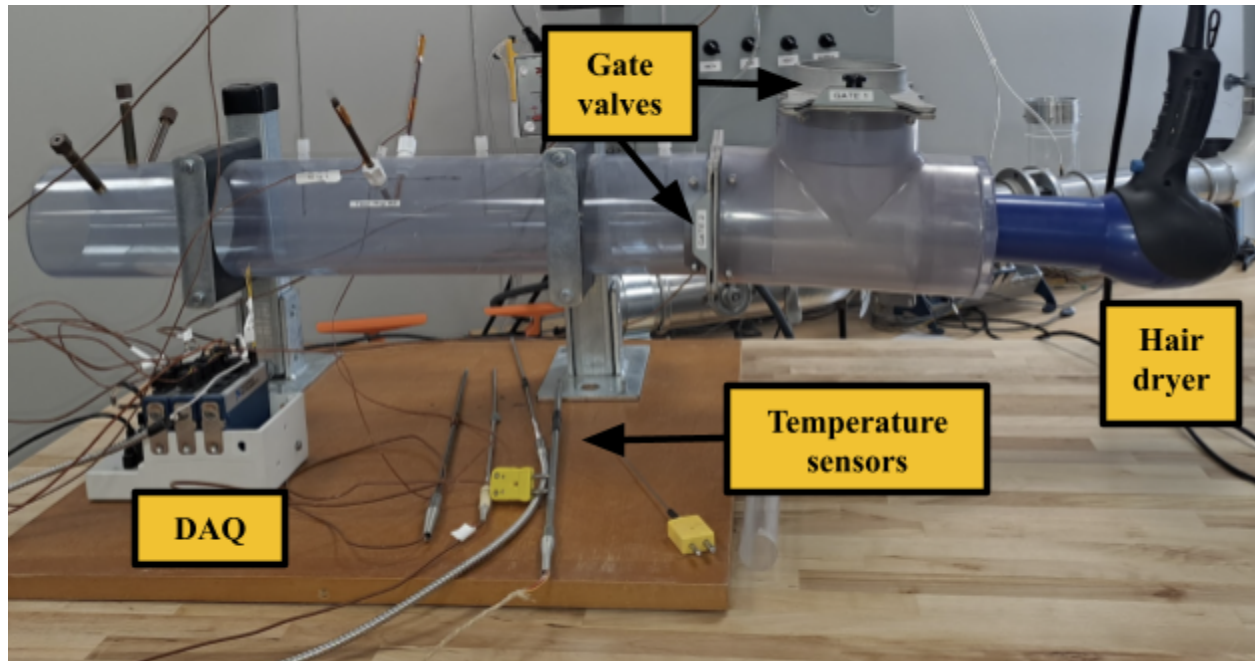


Figure 1. Test rig with annotations showing major components

2 Safety

Safety goggles must be worn at all times while operating the dynamic sensor test rig due to potential hazards. Flying debris and electrical shock are a threat during operation because the test rig uses blowing air and electrical heaters. During operation, neither gate valve should be closed at the same time because the hair dryer will overheat due to a lack of airflow. Additionally, the hardware used for the rig may fail, so it is important to be aware of the surroundings and inspect the hardware for any potential hazards before and after operation.

3 Hair Dryer Operation

There are five operational modes, which are combinations of the hair dryer's settings. The first operational mode is a low temperature setting and a low fan speed setting. The second mode is the low temperature

setting with high fan speed. The third mode is high temperature with low fan speed. The fourth mode is high temperature with high fan speed. These modes are activated by flipping the fan and heat settings switches on the hair dryer. The fifth operational mode is a cool-down, where the heater is off, but the fan still runs. This mode is activated by the blue button. For testing, the fourth operational mode will be used, where the hair dryer runs at high fan speed and high temperature. Figure 2 shows the hair dryer in more detail.



Figure 2. Close-up of hair dryer component of test rig

4 Data Collection

The LabView user interface is located in the C Drive in the folder AHU with the name 'Temperature Main 0.8'. After clicking on the file to open the software, the interface shown in Figure 3 will appear. The legend shows sensor readings of temperature in units of Celsius and voltage in units of millivolts. The acquired Data section shows the sensor readings in real time in the form of a graph.

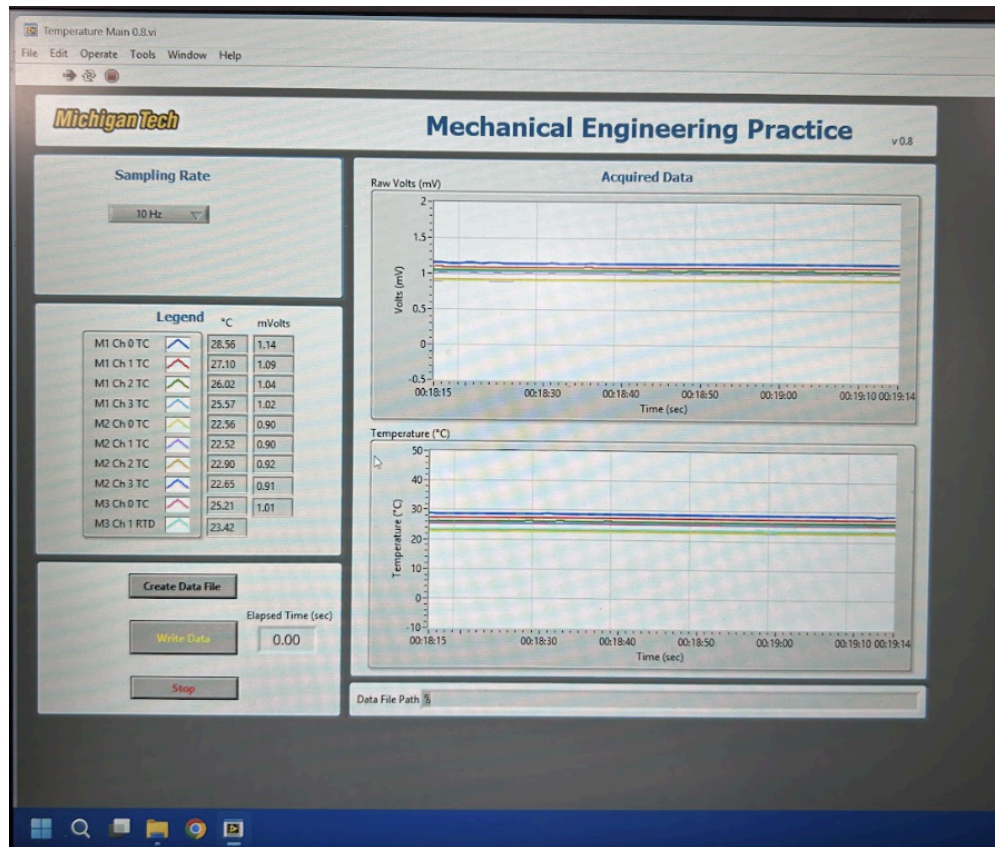


Figure 3. Lab View user interface after opening

To create the data file, click on 'Create Data File', and when prompted, choose to save the data file in a location where it can be easily found so that the data can be plotted for verification. By default, the data will be saved to a folder located in the C Drive. Click on 'Write Data' to start collecting data, and to save the data, click on 'Write Data' again. Clicking on 'Stop' will stop the LabView program [1].

To collect data that will allow for analyzing temperature sensor response, a step change in temperature must be mimicked using hot air flow from the hair dryer. To do this, the temperature sensor must be inserted into the test rig after the sensor is reading a constant room temperature and the hair dryer has had time to heat up. Figure 4 shows a temperature sensor properly inserted into the test rig.

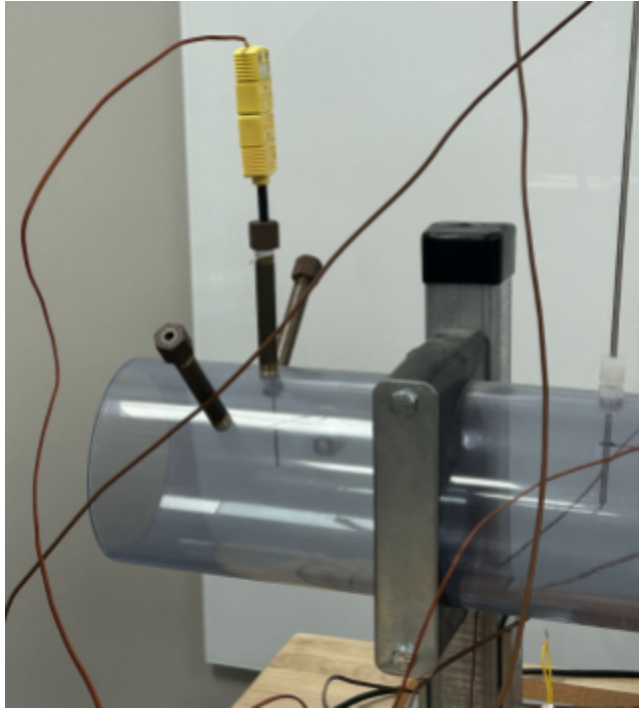


Figure 4. Temperature sensor inserted into test rig

5 Testing Considerations

In an emergency situation, the rig can be shut down by turning off the hair dryer or by unplugging the rig from the wall entirely to stop all flow of electricity through the rig.

Remember that the temperature sensors are sensitive, meaning they will pick up body heat transferred from people nearby and from touching the sensor. This added heat will create a bias in the collected data because the initial temperature reading will not be at room temperature.

The software does not specify which sensor is being recorded so it is important to trace the sensor to the channel it is plugged into on the DAQ. Then, make note of each test run and its corresponding sensor, to eliminate confusion when analyzing data.

6 Troubleshooting Tips

The hair dryer may stop unexpectedly due to thermal overload of the circuit breaker located in the plug. There is a push button, integral to the electrical plug, for resetting the breaker. [2]

Critical data should not be taken immediately. The DAQ may need to warm up for at least 10 minutes before it will be ready to start writing data.

7 References

[1] “ME2911-SOP-TemperatureRig”, in ME2911 canvas course, last accessed on 02/11/2026

[2] “ME2911.W5.TA”, in ME2911 canvas course, last accessed on 02/11/2026